

SAM10 MIDTERM PROJECT — AWS ARCHITECTURE DIAGRAM

Deliverable 2: 1-Page AWS Architecture Diagram Blueprint | MediCare Direct System

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System Name: MediCare Direct	AWS Target Region: ap-southeast-1 (Singapore)

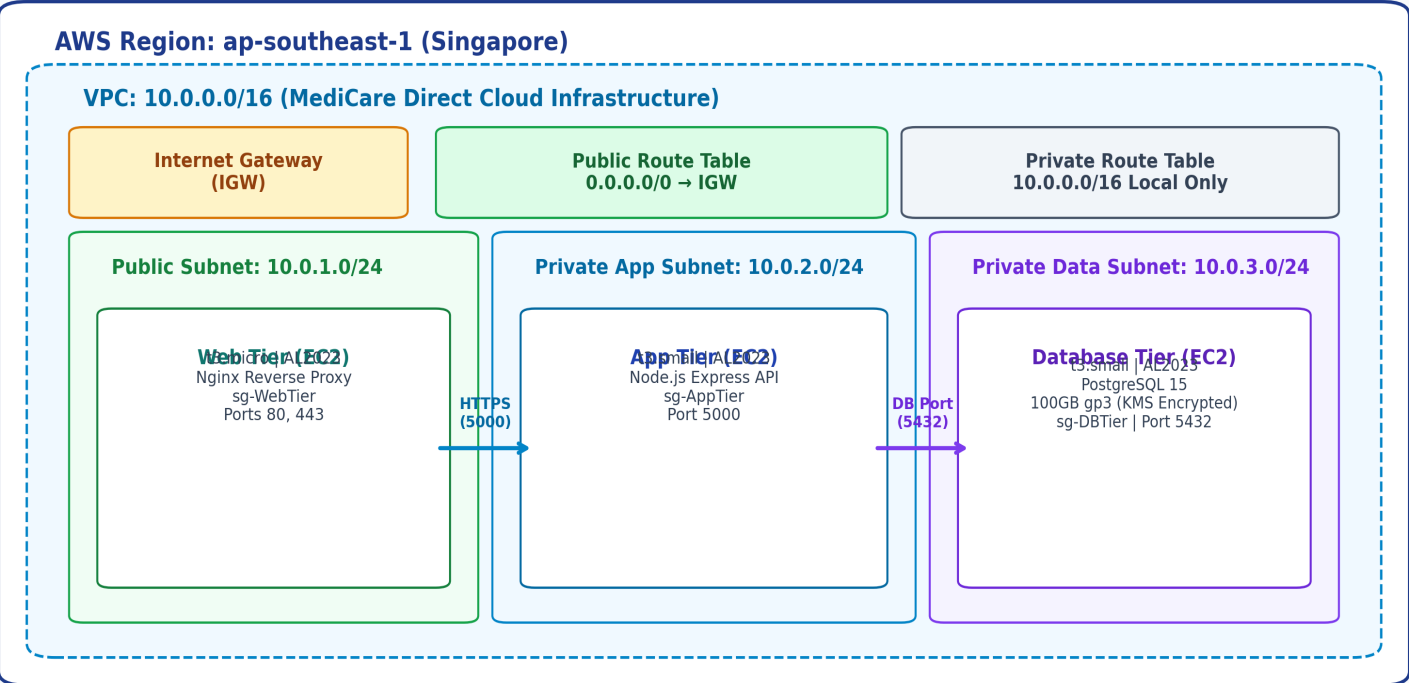


Figure 1: Production AWS 3-Tier Cloud Infrastructure Diagram for MediCare Direct System.

ARCHITECTURE COMPONENT SPECIFICATION & TRAFFIC PATHS

AWS Infrastructure Component	Network Location / CIDR	Security & Operational Function
Internet Gateway (IGW)	VPC Boundary (10.0.0.0/16)	Enables public Internet connectivity for Web Tier; attached to Public Route Table.
Public Subnet (Web Tier)	10.0.1.0/24 (Public Subnet)	Hosts t3.micro EC2 NGINX web proxy in sg-WebTier; terminates HTTPS 443.
Private App Subnet (API Tier)	10.0.2.0/24 (Private Subnet)	Hosts t3.small EC2 Node.js API in sg-AppTier; accepts Port 5000 only from sg-WebTier.
Private Data Subnet (DB Tier)	10.0.3.0/24 (Private Subnet)	Hosts t3.small EC2 PostgreSQL DB in sg-DBTier; accepts Port 5432 only from sg-AppTier.
AWS IAM & SSM Session Mgr	AWS Cloud Management	Governs administrative access; disables public SSH Port 22 in favor of SSM Session Mgr.